```c
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>

#define SIZE 5
#define ITEMS 10

int buffer[SIZE];
int in = 0, out = 0;

sem_t empty, full;
pthread_mutex_t lock = PTHREAD_MUTEX_INITIA

void *producer(void *arg) {
    for (int i = 1; i <= ITEMS; i++) {
        sem_wait(&empty);
        pthread_mutex_lock(&lock);

        buffer[in] = i;
        printf("Produced: %d\n", i);
        in = (in + 1) % SIZE;

        pthread_mutex_unlock(&lock);
        sem_post(&full);
    }
    return NULL;
}

void *consumer(void *arg) {
    for (int i = 1; i <= ITEMS; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&lock);

        int item = buffer[out];
        printf("Consumed: %d\n", item);
        out = (out + 1) % SIZE;
```

```
        sem_post(&empty);
    }
    return NULL;
}

int main() {
    pthread_t p, c;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&lock);

    printf("Producer-consumer completed.\n");
    return 0;
}
```

```
 GNU nano 7.2
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

pthread_mutex_t resource1 = PTHREAD_MUTEX_INITIALIZER;
pthread_mutex_t resource2 = PTHREAD_MUTEX_INITIALIZER;

void *thread1(void *arg) {
    pthread_mutex_lock(&resource1);
    printf("Thread 1 acquired Resource 1\n");
    pthread_mutex_lock(&resource2);
    printf("Thread 1 acquired Resource 2\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);
    return NULL;
}

void *thread2(void *arg) {
    pthread_mutex_lock(&resource1);
    printf("Thread 2 acquired Resource 1\n");
    pthread_mutex_lock(&resource2);
    printf("Thread 2 acquired Resource 2\n");

    pthread_mutex_unlock(&resource2);
    pthread_mutex_unlock(&resource1);
    return NULL;
}

int main() {
    pthread_t t1, t2;

    pthread_create(&t1, NULL, thread1, NULL);
    pthread_create(&t2, NULL, thread2, NULL);

    pthread_join(t1, NULL);
    pthread_join(t2, NULL);
```

```c
    pthread_join(t1, NULL);
    pthread_join(t2, NULL);

    printf("Completed without deadlock.\n");

    pthread_mutex_destroy(&resource1);
    pthread_mutex_destroy(&resource2);
    return 0;
}
```

```
      |                  ~~~~~~~^~~
sravya@Sravya1955:~/os$ ./p12a
Produced: 1
Produced: 2
Produced: 3
Produced: 4
Produced: 5
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
Consumed: 5
Produced: 6
Consumed: 6
Produced: 7
Consumed: 7
Produced: 8
Produced: 9
Produced: 10
Consumed: 8
Consumed: 9
Consumed: 10
Producer-consumer completed.
sravya@Sravya1955:~/os$ nano p12b.c
sravya@Sravya1955:~/os$ gcc -Wall -Wextra p12b.c -o p12b -pthread
p12b.c: In function ‘thread1’:
p12b.c:8:21: warning: unused parameter ‘arg’ [-Wunused-parameter]
    8 | void *thread1(void *arg) {
      |               ~~~~~~~^~~
p12b.c: In function ‘thread2’:
p12b.c:19:21: warning: unused parameter ‘arg’ [-Wunused-parameter]
   19 | void *thread2(void *arg) {
      |               ~~~~~~~^~~
sravya@Sravya1955:~/os$ ./p12b
Thread 1 acquired Resource 1
Thread 1 acquired Resource 2
Thread 2 acquired Resource 1
Thread 2 acquired Resource 2
Completed without deadlock.
sravya@Sravya1955:~/os$
```